Enterprise Agentic RAG Architecture

Bulk ingestion, permission-aware retrieval, bounded agent loop, grounded answers, and operational controls

1 BULK INGESTION | OFFLINE / ASYNCHRONOUS

Batch folder or object-store import; evaluation JSON stays outside the knowledge index

Document batch
PDF files + role manifest validate type, size, checksum

→ **Parse and validate ACL**
Docling layout + PyMuPDF pages Preserve tables, image assets, OCR/captions

→ **Chunk with provenance**
Table-aware text chunks + image records Keep rows, captions, page, filename, ACL

→ **Nomic embeddings + graph**
Text v1.5 for text/tables; Vision v1.5 for images Shared vector space + Docling-Graph relations

→ **Persist searchable data**
Neo4j vector + full-text indexes Tables, images, graph, ACLs + provenance

Bulk jobs are idempotent: validate first, process in batches, record status, and re-index changed documents by version/checksum.

2 ONLINE QUESTION FLOW | SYNCHRONOUS REQUEST PATH

Every retrieval round uses server-derived permissions

User
Question + authenticated session

→ **Streamlit UI**
Chat interface; no trusted role supplied by client

→ **FastAPI auth + RBAC**
Validate JWT; derive user + role; issue session_id

→ **Bounded agent controller**
Use session turns; decompose query; track evidence and budgets

→ **Authorization gate**
Get eligible document IDs from Neo4j ACL metadata before search

Session and traffic controls
Server-issued session_id; history scoped to user + role with idle TTL (SQLite locally / Redis at scale). Enforce max turns/session, concurrent-session cap, and per-user/session rate limits.

Vector retrieval
Neo4j vector index Text query searches text + images

Keyword retrieval
Neo4j full-text index exact terms and identifiers

Graph retrieval (optional)
Depth-capped; enforce ACL at each edge

Merge, deduplicate, rerank
RRF / measured score fusion; retain source + page + ACL

Evidence sufficient?

YES → **Context assembly**
Recheck ACL; include only authorized evidence

NO: targeted retrieval, max 3 rounds

NO: abstain when budget ends

LLM answer generation
Use evidence only; include source IDs and pages

Verify and respond
Check claim support; cite filename + page; abstain if weak

Insufficient information
Return a clear abstention; do not invent an answer

CROSS-CUTTING ENTERPRISE CONTROLS

Security
ACL before each search + before LLM; tenant scope; secrets; audit events; PII safeguards.

Reliability
Request timeout, bounded hops/tokens, retries for transient faults, idempotent ingestion.

Observability
Latency, errors, retrieval IDs/scores, model/version, abstention, cost; redact content.

Evaluation
Recall@k, citation/page accuracy, faithfulness, unknown-question abstention, ACL leakage = 0.

Design target: Neo4j for vector + graph retrieval | Preserve row structure, images, ACL, and page provenance